# Methodological and Controller Enhancements to Predictive Speed and Separation Monitoring via Robust Control Barrier Functions

**Antigravity & HRC Research Team**

November 6, 2026

## Abstract

Key enhancements include: (1) transitioning from discrete point-to-point keypoints to continuous volumetric swept-sphere capsules (SSVs) using Christer Ericson's segment-to-segment distance algorithm with continuous Jacobian interpolation; (2) establishing the formal algebraic identity linking the predictive stopping-horizon minimum $d_{\min}$ with the normative ISO 10218 $S_p$ requirement while correcting the scalar braking deceleration $a_s > 0$; (3) restoring uniform relative degree 1 by establishing a lower bound $\tau \in [\Delta t, T_f]$ and eliminating non-smooth ReLU velocity penalty terms; (4) deriving the exact closed-form analytical drift Lie derivative $L_f h(\chi)$ with line-of-sight projector operators; (5) resolving the sign inversion and severe dimensional mismatch ([m] vs. [m/s]) of the disturbance term via an ellipsoidal state-dependent covariance quadratic form $\sigma_{\text{eff}} = \sqrt{\nabla_\psi h \Sigma_\psi (\nabla_\psi h)^T}$; (6) converting the ill-conditioned hard joint-space tube into a convex, always-feasible Soft-Tube QP with slack variable $\varepsilon_k$ while preserving operational safety shields;

## 1. Introduction & Problem Context

Main limitations in the original paper:

1. **Geometric Occlusion & Intermediate Collisions:** Modeling agents via discrete 0D point clouds ($k = 7$ robot links, $m = 18$ human joints) left link cylinders and intermediate limbs unprotected.

2. **Heuristic Non-Smooth Barrier Term:** Adding an ad-hoc velocity penalty $\text{ReLU}(C_0 - d_{\min})\frac{v_r T_r}{C_0}$ degraded barrier differentiability.

3. **Relative Degree Degeneration:** Allowing the look-ahead horizon to include $\tau = 0$ caused the input-to-state Lie derivative to vanish ($L_g h = \mathbf{0}^T$), turning the differential inequality into an uncontrolled algebraic state constraint.

4. **Dimensional Inconsistency & Inverted Disturbance:** Adding $+h_\chi \Delta_H$ introduced an additive spatial offset in [m] directly to a velocity-rate equation [m/s], while the positive sign inadvertently loosened the barrier under worst-case disturbances.

5. **Hard Tube QP Infeasibility:** The hard trajectory tube constraint $|q_{k+1} - \bar{q}_{k+1}| \leq \delta_{\text{nom}}$ frequently clashed with safety deceleration requirements, causing QP failure and requiring ad-hoc recovery heuristics.

## 2. Kinematic & Volumetric Modeling

### 2.1. Continuous Swept-Sphere Capsule (SSV) Geometry

In the original manuscript, both human and robot were represented as discrete points:

$$\mathcal{H} = \{(p_{hi}, v_{hi}, \dot{v}_{hi})\}_{i=1}^{m}, \quad \mathcal{R} = \{(p_{rj}, v_{rj}, \dot{v}_{rj})\}_{j=1}^{k} \quad (1)$$

This discrete formulation is vulnerable to collisions occurring along the cylindrical shafts between keypoints (e.g., forearm, upper arm, torso).

In the revised methodology, robot links and human limbs are modeled as continuous 3D line segments inflated by bounding radii $R_r$ and $R_h$ (Swept-Sphere Volumes, SSVs). Let link $j$ be defined by the segment $S_{r,j} = [p^1_{r,j}, p^2_{r,j}]$ and human limb segment $i$ by $S_{h,i} = [p^1_{h,i}, p^2_{h,i}]$. Any point along the robot link and human limb is parameterized by curvilinear coordinates $s, t \in [0, 1]$:

$$p_r(s) = p^1_{r,j} + s(p^2_{r,j} - p^1_{r,j}) \quad (2)$$

$$p_h(t) = p^1_{h,i} + t(p^2_{h,i} - p^1_{h,i}) \quad (3)$$

The closest points $(p^*_r, p^*_h)$ between the two segments are found by solving the convex quadratic distance problem:

$$(s^*, t^*) = \arg\min_{s,t \in [0,1]} \|p_r(s) - p_h(t)\|_2^2 \quad (4)$$

We implemented this analytically via Christer Ericson's segment-to-segment algorithm (`segment_to_segment_distance`), resolving clamping conditions across all four boundary segments and internal voronoi regions. The net surface separation distance is:

$$d_{\text{surface}} = \|p_r(s^*) - p_h(t^*)\|_2 - (R_{r,j} + R_{h,i}) \quad (5)$$

The unit line-of-sight vector is defined along the optimal segment segment:

$$u_{rh} = \frac{p_r(s^*) - p_h(t^*)}{\|p_r(s^*) - p_h(t^*)\|_2} \quad (6)$$

Crucially, the effective translational Jacobian at the contact point $p_r(s^*)$ is obtained via convex linear interpolation of the endpoint link Jacobians:

$$J^*_{\text{lin}} = (1 - s^*) J_t(p^1_{r,j}) + s^* J_t(p^2_{r,j}) \quad (7)$$

Similarly, the contact Cartesian velocity is $v_r^* = (1 - s^*)v_{r,j}^1 + s^* v_{r,j}^2$. This ensures that the control action dynamically rotates and translates the actual volumetric collision boundary.

### 2.2. Rigorous Kinematics & Braking Deceleration Capability

In the original manuscript, $a_s$ was defined ambiguously with conflicting sign notations. In the revised formulation, $a_s \in \mathbb{R}_{>0}$ is defined strictly as the **scalar maximum braking deceleration capability** of the robot.

During an approach phase ($v_{r\|}^0 < 0$), braking is initiated after reaction time $T_r$. Deceleration applies an opposing acceleration $+a_s > 0$. The terminal stopping instant $T_f$ when the robot reaches zero approach velocity is:

$$T_f = T_r + \frac{|v_{r\|}^0|}{a_s} = T_r - \frac{v_{r\|}^0}{a_s} > T_r \tag{8}$$

### 2.3. Formal Identity with ISO 10218-2 ($S_p$)

We need to formally state what we told in the rejected paper: our $d_{min}$ formulation is equivalent to the ISO 10218 SSM one and the classical formulations represent a worst case approximation of our formulation. Idea:

**Theorem 1 (Equivalence of Stopping Horizon)** *Assume monotonic head-on human-robot approach ($v_{r\|}^0 < 0, v_{h\|}^0 > 0, a_{h\|} = 0$). Then the minimum distance predicted over the stopping horizon $[0, T_f]$ is attained at $T_f$ and satisfies:*

$$d(T_f) = d^0 - (S_h + S_r + S_s) \tag{9}$$

*Consequently, enforcing the predictive condition:*

$$\min_{\tau \in [0, T_f]} d(\tau) \geq C_0 \iff d^0 \geq S_p \tag{10}$$

*Proof.* Integrating relative velocity $\dot{d}(s) = v_{r\|}(s) - v_{h\|}(s)$ from 0 to $T_f = T_r + T_b$:

$$\begin{aligned} d(T_f) &= d^0 + \int_0^{T_r} (v_{r\|}^0 - \hat{v}_h) ds \\ &\quad + \int_{T_r}^{T_f} (v_{r\|}^0 + a_s(s - T_r) - \hat{v}_h) ds \\ &= d^0 + v_{r\|}^0 T_r - \hat{v}_h T_r + v_{r\|}^0 T_b + \frac{1}{2} a_s T_b^2 - \hat{v}_h T_b \end{aligned} \tag{11}$$

Substituting $T_b = -v_{r\|}^0 / a_s$ yields $v_{r\|}^0 T_b + \frac{1}{2} a_s T_b^2 = -\frac{(v_{r\|}^0)^2}{2a_s} = -S_s$. Collecting terms:

$$\begin{aligned} d(T_f) &= d^0 - |v_{r\|}^0| T_r - \hat{v}_h (T_r + T_b) - S_s \\ &= d^0 - (S_r + S_h + S_s) \end{aligned} \tag{12}$$

Setting $d(T_f) \geq C_0$ yields $d^0 \geq S_h + S_r + S_s + C_0 = S_p$. ■

This identity proves that classical SSM is a single, static evaluation point trapped at $\tau = T_f$. Our predictive barrier generalizes SSM: when human motion decelerates ($a_{h\|} < 0$), the true minimum occurs earlier ($\tau^* = \tau_2 < T_f$), reducing over-conservatism.

## 3. Predictive Robust Control Barrier Function Theory

### 3.1. Elimination of Heuristic Penalties & Relative Degree Restoration

The original paper defined the barrier function as:

$$h_{\text{orig}} := d_{\min} - C_0 + \text{ReLU}(C_0 - d_{\min}) \frac{v_r T_r}{C_0} \tag{13}$$

This formulation has two fundamental flaws:

1. **Loss of Relative Degree at $\tau^* = 0$:** The minimum distance function $d(\tau)$ evaluated at $\tau = 0$ is simply $d^0 = \|p_r - p_h\|$. Its gradient with respect to control action $\dot{v}_r$ vanishes:

$$L_g h = \frac{\partial d^0}{\partial v_r} = \mathbf{0}^T \tag{14}$$

   When $\tau^* = 0$ is selected as the minimizer, the system possesses relative degree 2. Enforcing a first-order CBF condition at $\tau^* = 0$ turns the inequality into an uncontrolled algebraic constraint, creating singular QP matrices.

2. **Non-Smooth Heuristic Perturbation:** The ReLU penalty introduced gradient discontinuities exactly along the boundary of the safe set.

**Rectified Barrier Formulation**

We define the canonical predictive barrier function:

$$h(\psi(\chi)) := d_{\min}(\psi(\chi)) - C_0 \tag{15}$$

with the evaluation domain restricted to the open look-ahead causal horizon:

$$\tau \in [\Delta t, T_f], \quad \Delta t > 0 \tag{16}$$

Because $\tau^* \geq \Delta t > 0$ for all control cycles:

$$L_g h = \nabla_\chi h \cdot g(\chi) = \tau^* u_{rh}^T \neq \mathbf{0}^T \tag{17}$$

This guarantees **relative degree 1** across the entire state space.

In alternative, we should define a High Order CBF to deal with cases where $\tau_{min} = 0$.

### 3.2. Ellipsoidal Robustness vs. Original Formulation

The original paper included uncertainty via the following constraint (Eq. 18):

$$L_f h + L_g h \mu + h_\chi(\chi) \Delta_H + \gamma h(\psi) \geq 0 \tag{18}$$

with $\Delta_H = \beta \sigma_p + \beta \sigma_v \bar{T}_f + \frac{1}{2} \beta \sigma_a \bar{T}_f^2$.

#### *3.2.1. Theoretical Flaws in the Original Form*

1. **Dimensional Inconsistency:** In Eq. (18), $L_f h$ and $L_g h \mu$ have units of velocity [m/s]. The term $\gamma h(\psi)$ has units $[1/\text{s}] \cdot [\text{m}] = [\text{m/s}]$. However, $h_\chi(\chi) = \frac{\partial h}{\partial \chi}$ has components of dimensionless order $\mathcal{O}(1)$ (or

[s]), multiplying $\Delta_H$ which has units of distance [m]. The resulting additive term has units of meters [m], violating dimensional homogeneity ([m/s] + [m] $\geq$ 0).

2. **Sign Inversion:** Adding $+h_\chi \Delta_H$ with positive sign increases the lower bound, effectively loosening the safety constraint when uncertainty $\Delta_H$ increases.

3. **State Independence:** The scalar $\Delta_H$ used a fixed horizon $\bar{T}_f$, ignoring whether the active collision threat is imminent ($\tau^* \to \Delta t$) or distant ($\tau^* \to T_f$).

#### *3.2.2. Rectified State-Dependent Covariance Quadratic Form*

Let $\Sigma_\psi = \text{diag}(\sigma_p^2, \sigma_v^2, \sigma_v^2, \sigma_a^2) \in \mathbb{R}^{4\times4}$ be the state uncertainty covariance matrix. Under Gaussian perception disturbance, the variance of the predicted distance margin is given by the quadratic form:

$$\sigma_{\text{eff}}^2(\tau^*) = \nabla_\psi h \, \Sigma_\psi \, (\nabla_\psi h)^T \tag{19}$$

Expanding this gives:

$$\sigma_{\text{eff}}(\tau^*) = \sqrt{\sigma_p^2 + (\tau^*)^2\sigma_v^2 + \frac{1}{4}(\tau^*)^4\sigma_a^2} \tag{20}$$

The parameter $\beta = 3.291$ enforces a 99.9% confidence bound. The robust barrier condition is formulated as:

$$\dot{h}(\psi) \geq -\gamma \left(h(\psi) - \beta\sigma_{\text{eff}}(\tau^*)\right) \tag{21}$$

Expanding into the QP inequality on joint acceleration $u$:

**Rectified Robust CBF Constraint**

$$\boxed{\begin{aligned}\tau^* u_{rh}^T J_t(q) u \geq &-\tau^* u_{rh}^T \dot{J}_t \dot{q} - L_f h \\ &- \gamma \left(h(\psi) - \beta\sigma_{\text{eff}}(\tau^*)\right)\end{aligned}} \tag{22}$$

This rectified form satisfies dimensional consistency and the uncertainty grows quadratically with the look-ahead horizon $\tau^*$, avoiding over-conservatism during close-proximity tracking.

### 3.3. Theoretical Guarantee for Non-Smooth Minimum Switching

Because $d_{\min} = \min_k d(\tau_k)$, the active minimizer $\tau^*$ switches discretely. The forward invariance of the non-smooth safe set $\mathcal{C}$ is formally guaranteed by the following theorem.

**Theorem 2** *Let $h_k(\psi) = d(\tau_k) - C_0$ for each candidate extremum $\tau_k \in \mathcal{T}$. If the control input $u$ satisfies:*

$$\dot{h}_{k^*}(\psi, u) \geq -\gamma(h_{k^*} - \beta\sigma_{\mathit{eff}}) \quad \textit{where } k^* = \arg\min_k h_k(\psi) \tag{23}$$

*then the safe set $\mathcal{C} = \{\chi \mid \min_k h_k \geq 0\}$ is forward invariant almost everywhere under Filippov solutions (Glotfelter et al., IEEE LC-SYS 2017, Proposition 2).*

Enforcing only the active minimum $k^*$ avoids the over-conservatism that would result from simultaneously bounding all non-critical prospective horizons.

## 4. Optimization Controller Architectures

#### *4.0.1. Soft-Tube Slack Formulation*

In the original manuscript, path deviation was constrained by a hard inequality:

$$|q_{k+1} - \bar{q}_{k+1}| \leq \delta_{\text{nom}} \tag{24}$$

Whenever an unexpected human intrusion required emergency deceleration, the safety CBF commanded rapid braking, while the hard tube demanded strict adherence to nominal path progression. These opposing constraints frequently rendered the QP primal infeasible.

In the rectified formulation, we introduce a non-negative slack variable $\varepsilon_k \in \mathbb{R}^n_{\geq 0}$:

$$-\delta_{\text{nom}} - \varepsilon_k \leq q_{k+1} - \bar{q}_{k+1} \leq \delta_{\text{nom}} + \varepsilon_k, \quad \varepsilon_k \geq \mathbf{0} \tag{25}$$

The augmented optimization problem is formulated as:

$$\begin{aligned}\min_{u_k, \ddot{\xi}_k, \varepsilon_k} \quad & \lambda_{\text{pos}}\|q_{k+1} - \bar{q}_{k+1}\|_2^2 + \lambda_{\text{vel}}\|\dot{q}_{k+1} - \dot{\bar{q}}_{k+1}\|_2^2 \\ & + \lambda_{\text{acc}}\|u_k\|_2^2 + \lambda_{\text{scaling}}(\dot{\xi}_{k+1} - 1)^2 + \lambda_{\text{tube}}\|\varepsilon_k\|_2^2 \\ \text{s.t.} \quad & q_{\min} \leq q_{k+1} \leq q_{\max}, \\ & \dot{q}_{\min} \leq \dot{q}_{k+1} \leq \dot{q}_{\max}, \\ & u_{\min} \leq u_k \leq u_{\max}, \\ & 0 \leq \dot{\xi}_{k+1} \leq 1, \quad \ddot{\xi}_k \leq \ddot{\xi}_{\max}, \\ & -\delta_{\text{nom}} - \varepsilon_k \leq q_{k+1} - \bar{q}_{k+1} \leq \delta_{\text{nom}} + \varepsilon_k, \\ & \varepsilon_k \geq \mathbf{0}, \\ & \text{Robust CBF Inequalities } \forall(i,j)\end{aligned} \tag{26}$$

Setting $\lambda_{\text{tube}} = 1.0 \times 10^5 \gg \lambda_{\text{pos}}$ guarantees that slack is engaged only when safety boundaries strictly prevent the robot from remaining within $\delta_{\text{nom}}$, guaranteeing mathematical QP feasibility at every control step.